

UNMASKING DEPRESSION:

USING ONLINE SOCIAL NETWORKS TO DETECT DEPRESSIVE SYMPTOMS

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A decorative background on the left side of the slide featuring a collage of social media icons including Pinterest, Facebook, YouTube, and Snapchat, set against a grey background with a pattern of small black dots.

LITERATURE OVERVIEW

- We read papers about how depression can be detected using different **OSNs (Online Social Networks)**.
- OSNs include: Facebook, Instagram, Twitter, etc.
- Our main reading is **Dr. Meeyoung Cha's "Activities on Facebook Reveal the Depressive State of Users" (2013)**, which focused on the correlation between depression and Facebook usage.
- This is a **Social Computing field**, an interdisciplinary field bridging technology and human behavior that offers innovative tools to understand and predict mental health trends.

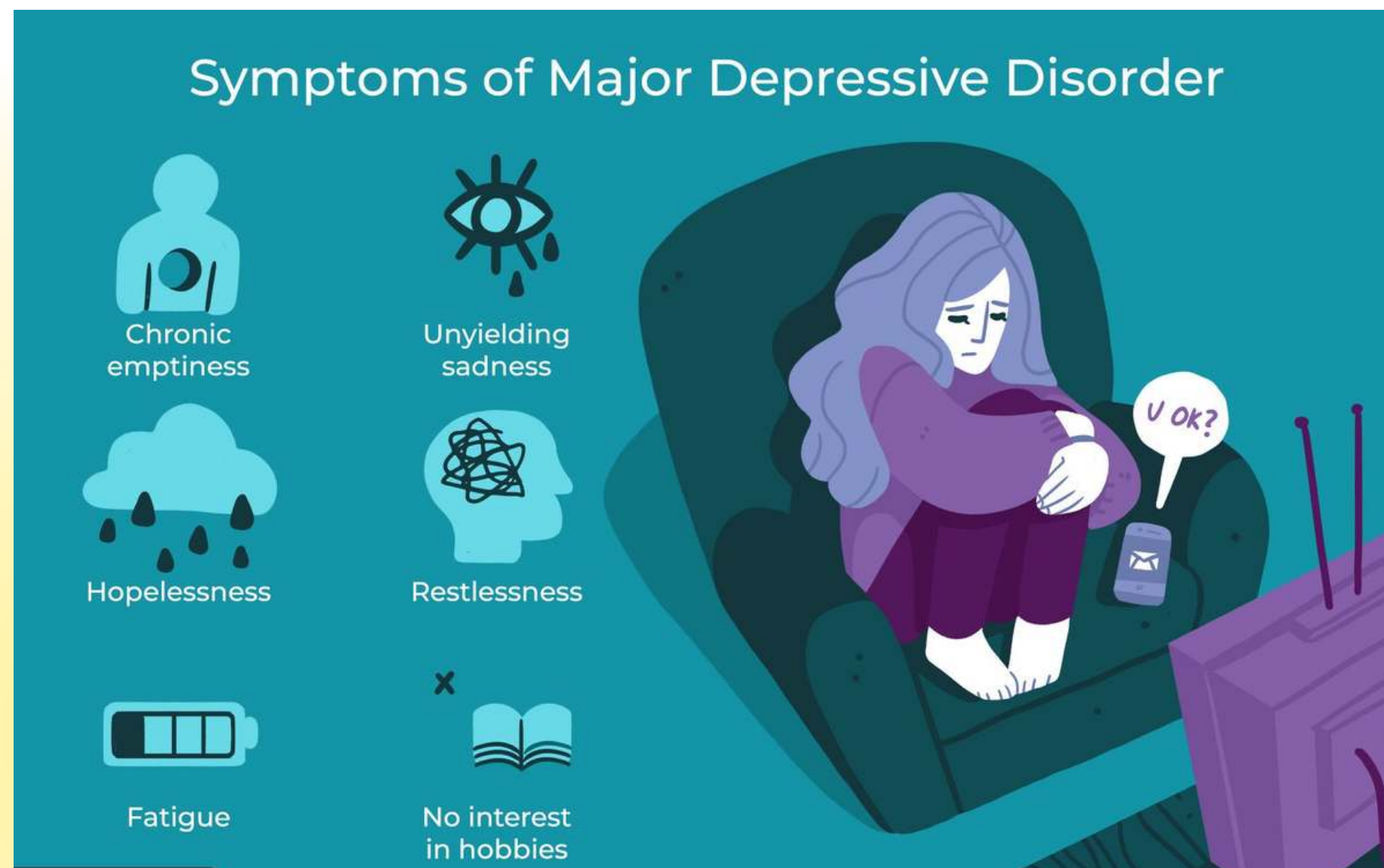


SCHOLAR INTRODUCTION

- Studies Social Computing, AI ethics, Online social networks, and Misinformation
- Professor at Max Planck Institute for Security and Privacy (MPI-SP) and KAIST
- Used to work as an assistant professor at Facebook



WHAT IS DEPRESSION?



Depression is one of the most common mental disorders, with a lifetime prevalence of 16.2% (Park et al., 2013), and it causes significant threats to individuals such as self-harm and suicide.

The WHO has emphasized the need for developing an interdisciplinary strategies to prevent suicide attempts through early risk identification (Malhotra & Jindal, 2022).

LITERATURE REVIEW

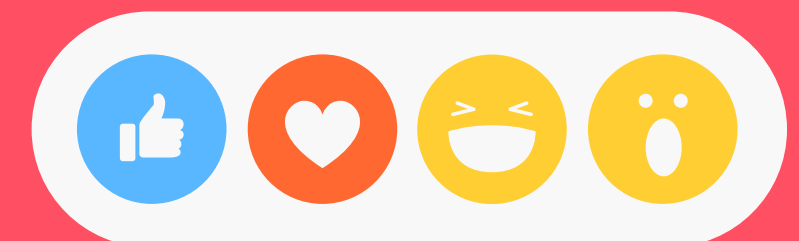
SUMMARY OF THE MAIN READING

- Summary of main reading:
 - + OSN patterns of 55 participants were analyzed both quantitatively and qualitatively
 - + Tested users' mental health status from a mobile application EmotionDiary on Facebook, which provided subjects with tips and facts about depression.
 - + Used traditional survey scales CES-D and BDI to check reliability
 - + Findings: depressed state manifests through having a lower number of friends on Facebook, less location tags and viewing more tips on EmotionDiary
- This paper proposed an innovative and interdisciplinary approach to detect depression in a nonclinical way, which is well suited for characteristics of depressed individuals.

LITERATURE REVIEW

THEORETICAL FRAMEWORK

- Emotional contagion theory explains how emotions spread through OSN, influencing users' mental health (Park et al., 2015).
- AI and machine learning theories focus on modeling human behavior and detecting emotional signals, emphasizing computational tools' potential in mental health diagnostics (De Choudhury et al., 2013; Malhotra & Jindal, 2022).



LITERATURE REVIEW

METHODOLOGIES



- Most common method: Questionnaires, in which the participants filled their demographic details, medical history, and the consent of letting the researchers observe their social media accounts.
- Less common: De Choudhury et al. (2021) and Reece and Danforth (2017) utilized Amazon's Mechanical Turk platform to crowdsource the subjects.
- Three of the papers conducted their research through developing web applications such as EmotionDiary in Cha's works and IBM Watson, a chatbot that cooperates with ChatGPT in Indo et al. (2024).
- Most of the studies used CES-D, Center for Epidemiological Studies Depression; the questionnaires included 20-item self-report scale to numerically find out the level of depression.

LITERATURE REVIEW

KEY THEMES

- Studies used social media as a predictive tool for mental health conditions.
- Researchers have consistently explored how users' online behaviors—such as linguistic patterns, posting frequency, and visual content—correlate with emotional states.
(Park et al., 2013; Reece & Danforth, 2017).
- A clear trend is the adoption of advanced machine learning techniques to automate the identification of depressive markers in large-scale datasets.
(De Choudhury et al., 2013; Malhotra & Jindal, 2022).
- Platform-specific differences also emerge as a focus, with studies comparing the effects of Facebook, Instagram, and Twitter on mental health.
(Limniou et al., 2022; Park et al., 2012).
- Recently, the field has shifted toward actionable applications, such as using social media for therapeutic interventions, as seen in studies on expressive writing and AI chatbots.
(Lee et al., 2016; Indo et al., 2024).



LITERATURE REVIEW

MAJOR FINDINGS

- 1** Negative linguistic patterns, reduced interactions, and altered posting behaviors are strong predictors on platforms like Facebook and Twitter (Park et al., 2013; Park et al., 2012).
- 2** On Instagram, however, visual markers, such as darker photo tones and lower vibrancy, are associated with depressive states (Reece & Danforth, 2017).
- 3** Advanced machine learning models demonstrate high accuracy in identifying depression and suicide risks from social media data (Malhotra & Jindal, 2022; De Choudhury et al., 2013).
- 4** Online interventions, such as expressive writing on Facebook and AI-driven chatbots, show promise in mitigating depressive symptoms (Lee et al., 2016; Indo et al., 2024).

LITERATURE REVIEW

IMPLICATIONS

- Studies demonstrate the feasibility of detecting depression symptoms and episodes using OSN , which can be valuable in clinical research.
- Findings can contribute to raising social awareness about mental health and promoting early intervention strategies.
- The development of online social network (OSN)-based healthcare support systems or services, like EmotionDiary, could significantly advance the mental health care field.



LITERATURE REVIEW

RESEARCH GAP

1

Small number of participants and a short experimental duration

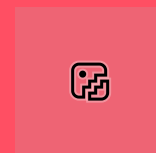


3

Most research does not adequately address diversity between age groups or gender

2

Findings are often constrained to specific platforms



4

Lack of ethical considerations, data privacy, and the potential misuse of predictive technologies



STUDY DESIGN

BACKGROUND KNOWLEDGE



DEPRESSION IN MONGOLIA:

- 44.6% percent of Mongolian population is under the age of 24.
(Davaasambuu, 2018).
- Anxiety, depression, and brain overwork are a significant problem in the general population of Mongolia.
(Lkhagvasuren, 2024).
- Mongolia has among the highest rates of adolescent suicide in East Asia and the Pacific.
(UNICEF, 2022).
- There's a significant lack of research regarding mental health of Mongolian citizens, let alone the youth.
(Davaasambuu, 2018).



STUDY DESIGN

RESEARCH OBJECTIVES

OBJECTIVES:

- Using emoji icons to determine depression among Mongolian students who are frequent OSN users
- Expand on the objectives of our initial chosen research by Dr. Cha
- Analyze the mental health state of Mongolian students
- Find out if some emojis are used opposite to their initial depiction in Gen-Z individuals

TIE TO LITERATURE REVIEW:

- Using OSN as a tool to determine depression (all research papers)
- Reapplying CES-D measures to detect depression (most papers)
- Being backed by an analysis made by a professional psychiatrist

ADDRESSED

LIMITATIONS:

- Targeting various social media
- Assuming equal gender proportion
Trying not to violate user privacy



THEORETICAL FRAMEWORK & HYPOTHESIS

Symbolic Interactionism - Emojis as a form of symbolic communication convey emotions and can reflect psychological states.

The frequency and type of emoji usage on OSNs by university students in Mongolia can predict depressive symptoms.



STUDY DESIGN

RESEARCH METHODOLOGY

STAGE 1: FINDING PARTICIPANTS

Sending out research invitations to students in universities in Mongolia to evaluate their mental health state (depression) with the help of professionals

- calls for frequent emoji users on various OSN
- limited to 100 participants
- compensation equivalent to 30RMB ~ 15'000MNT

STAGE 2: QUESTIONNAIRE I - DATA COLLECTION

- Ask what OSN the subject uses the most
- Ask 5 most frequently used emoji icons of the subject

STAGE 3: DATA SELECTION

- Out of 500 emojis, determine the most commonly used 20 emojis using statistical tools (more than 20 emojis would require too much time commitment from the subject for questionnaire II)

STUDY DESIGN

RESEARCH METHODOLOGY

STAGE 4: QUESTIONNAIRE II - INTERPRETATION

- Determine the usage purpose of the chosen 20 emoji icons using a scale of 1 to 5 with 1 indicating “depressed” and 5 indicating “cheerful” states.
- Allow participants to give additional context for their emoji use, especially if some emojis might be used ironically

STAGE 5: PRE- ANALYSIS

- Subjects with frequent use of “depressed” emoji icons (cross-referencing the data from Questionnaire I) will be put in the initial prediction of individuals with depressive symptoms

STAGE 6: INTERVIEW

- Detect depression through CES-D measures
- Interview with a professional psychiatrist

STAGE 7: FINAL ANALYSIS

- Determine whether our hypothesis aligns with the results from professional diagnosis
- Conclude the research



STUDY DESIGN

EVALUATION AND IMPLICATIONS

LIMITATIONS

- Small sample size due to limited budget
- Reliance on self-reported data
- Focusing on specific age-group
- Inapplicable to other cultures

FUTURE DIRECTIONS

- Expansion of scope and duration of research through more budget.
- Implementing a similar study in other cultures/countries and age range
- Determining the emojis that resemble depression through AI

STRENGTHS & IMPLICATIONS

- Localized insights - provide culturally relevant insights into emoji usage and mental health correlations
- Ideal demographic - university students are likely to use emojis extensively on OSNs
- Uniformity in emoji interpretation - there wouldn't be different perceptions of emojis/symbolism due to the subjects being from the same background
- Academic contribution & Policy Impact - entry point to discuss and address mental health issues in Mongolia



STUDY DESIGN



FEASIBILITIES

A TOTAL BUDGET OF 3250RMB

- 3000RMB - Compensation of 30RMB for each participant in our research
- 250RMB - Compensation of 250RMB for our voluntary psychiatrist



RESPONSIBILITY OF RESEARCH GROUP MEMBERS:

Khatanbuuvei - Preparation and Communication

- Preparing and distributing research invitations to students.
- Coordinating with the psychiatrist for interviews and evaluations.

Namuunzaya - Questionnaire Design and Data Collection

- Drafting and refining Questionnaire I and II.
- Collecting responses from participants.
- Ensuring the data is well-organized and complete.

Undran - Data Analysis and Technical Assistance

- Identify the top 20 emojis from Questionnaire I.
- Analyzing results from Questionnaire II and preparing pre-analysis data.
- Technical assistance

Nemuulen - Evaluation and Final Analysis

- Facilitating professional evaluations with the psychiatrist.
- Cross-referencing psychiatric results with pre-analysis findings.

DISCUSSION

1. Do you think using Social Media (OSN) to detect depression symptoms is better than the traditional ways?
2. What are your thoughts on the privacy and ethical concerns associated with using online social networks to detect depression?
3. Would you be willing to participate in a study where your social media activity is analyzed to detect signs of depression? Why or why not?
4. How much confidence would you have in the accuracy and reliability of the results of the study based on your social media activity? Why?
5. Do you think the use of particular emojis can be an indicator of a person's mental state?
6. Do you tend to use "sad" emojis when you're feeling down?
7. Which emojis can have opposite meaning to their original depiction? (For example: Is the loud crying emoji  really used when people try to convey the meaning of laughter?)

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**THANK YOU FOR
YOUR ATTENTION!**